

Q1. Predictive Parser Design (Concept mapping) 192424325

ZONE 1: The grammar

$$E \longrightarrow TE'$$

$$E' \longrightarrow TE' \mid \epsilon$$

$$T \longrightarrow FT'$$

$$T' \longrightarrow * FT' \mid \epsilon$$

$$F \longrightarrow (\epsilon) \mid id$$

The grammar
is left recursive,
and properly left
factored

Zone 2:

First Sets

$$\text{First}(E) = \{ (, id \}$$

$$\text{First}(E') = \{ +, \epsilon \}$$

$$\text{First}(T) = \{ (, id \}$$

$$\text{First}(T') = \{ *, \epsilon \}$$

$$\text{First}(F) = \{), \$ \}$$

FOLLOW SETS

$$\text{FOLLOW}(E) = \{), \$ \}$$

$$\text{FOLLOW}(E') = \{), \$ \}$$

$$\text{FOLLOW}(T) = \{ +,), \$ \}$$

$$\text{FOLLOW}(T') = \{ +,), \$ \}$$

$$\text{FOLLOW}(F) = \{ *, +,), \$ \}$$

Zone 3 :

Non Terminals

ID + * () \$

E

$E \rightarrow TE'$

$E \rightarrow TE'$

E'

$E \rightarrow TE'$

$T \rightarrow FT'$

$E \rightarrow +TE'$

T

$T \rightarrow FT'$

T'

$T \rightarrow FT'$

$T' \rightarrow \epsilon$

F

$F \rightarrow id$

$F \rightarrow (+)$

Zone 4 :

Stack

I.B

P. Action

\$ E

id + id * id \$

$E \rightarrow TE'$

\$ E' T

id + id * id \$

$T \rightarrow FT'$

\$ E' T' F

id + id * id \$

$F \rightarrow id$

\$ E' T' id

\$ E' T'

\$ E'

\$ E' T +

\$ E' T' F

\$ E' T' id

\$ E' T'

\$ E' T' F *

\$ E' T' F

\$ E' T' id

\$ E' T'

\$ E'

id + id * id \$

+ id * id \$

+ id * id \$

id + id \$

id * id \$

id + id \$

* id \$

* id \$

id \$

id \$

\$

\$

match id

$T' \rightarrow \epsilon$

$E' \rightarrow T E'$

Match +

$T \rightarrow F T'$

match id

$T' \rightarrow * F T'$

match *

$F \rightarrow id$

match id

$T' \rightarrow \epsilon$

$E' \rightarrow \epsilon$

Snippet OF

PREDICTIV PARSER

Class Predictive parser :

```
def __init__ ( self , start ) :
```

```
    self . elements = [ $ , start ]
```

```
def push ( self , production ) :
```

```
    self . element . extend ( reversed ( prod ) )
```

```
def match ( self , current_token ) :
```

```
    if self . elements [ -1 ] == current_tok :
```

```
        return self . element . pop ()
```